

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #14 - Healthcare & Telemedicine - Corporate Mental Wellness & Counseling Platform

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1. Scenario and approach

Scenario: An anonymous counselling portal for a company. Employees can check their stress levels, book confidential sessions with a licensed therapist and follow their own wellness indicators, and at no point does the employer learn who used it.

This report covers Lab 1 for Problem Statement #14. It works from the given scenario through to a set of requirements, a use-case model and four use-case flows.

The problem statement fixes the goal and names two stakeholders, but it leaves the mechanism open. It does not say what the questionnaire contains, how the pseudo-ID is produced, or where the link between an employee and their pseudo-ID is kept. Section 2 settles those points before anything depends on them, so every acceptance criterion has something concrete to test against.

The counts follow the handout: five functional requirements and two non-functional ones, each with a measurable pass and fail condition, at least three actors, and at least one include and one extend relationship in the diagram.

2. Stated assumptions

Assessment instrument. The questionnaire is ten items, each answered on a five-point frequency scale (Never 0, Rarely 1, Sometimes 2, Often 3, Very Often 4), giving a total out of 40. Bands are low (0-10), moderate (11-20), high (21-30) and severe (31-40). This is a wellness self-assessment for triage and self-awareness. It is not a diagnostic instrument and the platform does not present it as one.

Two separate identities. The employee has a corporate identity (employee ID and corporate email) used once, to prove they are entitled to the benefit. Everything on the counselling side is keyed on a pseudo-ID such as WEL-7F42A91C, generated at random rather than derived from the corporate identity, so it cannot be reversed.

Where the link is kept. The pseudo-ID to employee mapping lives in an identity store separate from counselling records. Booking, assessment and notes data hold the pseudo-ID and nothing else, so a leak of the counselling database does not identify anybody.

Out of scope for Lab 1. Session delivery (the video or chat call itself), therapist licensing checks and payment or billing are outside this iteration.

3. Requirements

Req ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall verify an employee against the corporate directory once at registration, then issue a randomly generated pseudo-ID, and shall use only that pseudo-ID in every record and screen available to a therapist, so that no corporate email address, employee ID or name is disclosed.	High	Pass: after an employee books a session, the therapist dashboard and the stored booking record show only the pseudo-ID in the format WEL-XXXXXXX, and the identity mapping is unreadable from the counselling database. Fail: any booking metadata field contains the employee's email, employee ID or full name, or the pseudo-ID can be derived from the corporate identity.	The whole service depends on this. An employee who thinks HR can see their name simply will not book.
FR-002	Functional	The system shall let an employee complete a ten-item stress self-assessment, each item answered on a five-point frequency scale, and shall return the total score out of 40 with its band (low, moderate, high or severe) once the final item is answered.	High	Pass: submitting a fully answered questionnaire returns a score between 0 and 40 and the band matching the published ranges, within 2 seconds, labelled as a wellness check rather than a diagnosis. Fail: no result is returned, the band contradicts the score, or a partly answered form is scored.	Gives the employee and the therapist the same starting point. The score is also what FR-005 checks against.
FR-003	Functional	The system shall allow an employee to book a counselling session only in a slot the therapist has published as free, and shall mark that slot unavailable in the same transaction.	High	Pass: when two employees confirm the same slot at the same moment, one booking is confirmed and the other receives a 'slot no longer available' message. Fail: both bookings are confirmed, or a slot outside the therapist's published availability is accepted.	Stops double booking. A therapist who finds two people waiting for the same slot has to turn one of them away.
FR-004	Functional	The system shall show an employee a trend of their own wellness indicators (stress score, sleep hours and sessions attended) across their six most recent assessments.	Medium	Pass: an account with six or more assessments plots all six in date order; an account with fewer plots only the assessments that exist. Fail: the chart shows another account's data, or fills dates that have no assessment with estimated values.	A trend is easier to read than a list of old scores, and it gives the employee a reason to keep taking the assessment.
FR-005	Functional	The system shall, whenever an assessment falls in the severe band, display regional crisis helpline details on the results screen and offer the earliest available priority appointment slot.	High	Pass: a submission scoring in the severe band shows the helpline panel and lists at least one priority slot starting within the next 24 hours. Fail: a severe score is presented as an ordinary result, with no helpline and no priority slot.	A severe score needs attention the same day, not after a two-day wait in the normal booking queue.
NFR-001	Nonfunctional (Performance & Security)	The appointment scheduler shall convert every slot to the viewer's local timezone correctly, including across daylight saving transitions, and shall return a slot list in under 2 seconds at peak load.	High	Pass: a benchmark of 500 concurrent users across IST, GMT and PST keeps 95th percentile slot-list latency below 2 seconds, and a booking made on a daylight saving changeover date shows the same absolute time to both parties. Fail: the two views disagree on the time, or 95th percentile latency exceeds 2 seconds.	Employees and therapists sit in different countries and book across timezones daily. An hour's error means somebody misses the session.
NFR-002	Nonfunctional (Security & Privacy)	The system shall hold the pseudo-ID to employee mapping in a store separate from all counselling data, shall encrypt that mapping, session notes and assessment answers with AES-256 at rest and TLS 1.3 in transit, and shall release an aggregate wellness report to a Wellness Program Administrator only where the reported group contains at least ten employees.	High	Pass: a dump of the counselling database yields no readable notes and no corporate identifier, and a report requested for a department of nine employees is refused with a 'group too small' message. Fail: plaintext notes appear in storage or logs, the mapping is readable from counselling data, or a report is released for a group smaller than ten.	Session notes are clinical records, and the mapping is the one table that would undo every pseudo-ID at once. Reports on very small teams would also let HR work out who has been using the service.

4. Actors

Actor	Kind	Role in the system
Corporate Employee	Primary	Takes assessments, books and attends sessions, tracks their own indicators. Known to the counselling side of the system only by a pseudo-ID.
Licensed Therapist	Primary	Publishes availability, conducts sessions and records notes. Sees the pseudo-ID, never the employee's corporate identity.
Wellness Program Administrator	Primary	HR-side owner of the programme. Reads aggregate uptake and wellness reports only, subject to the group size floor in NFR-002.
Corporate Identity Provider	Supporting	The employer's existing directory or SSO. Confirms once that the person is a current employee, and is never told what they do afterwards.
Crisis Support Service	Supporting	External helpline the platform hands off to when an assessment scores in the severe band.
Notification Service	Supporting	External email and push provider that delivers booking confirmations and reminders.

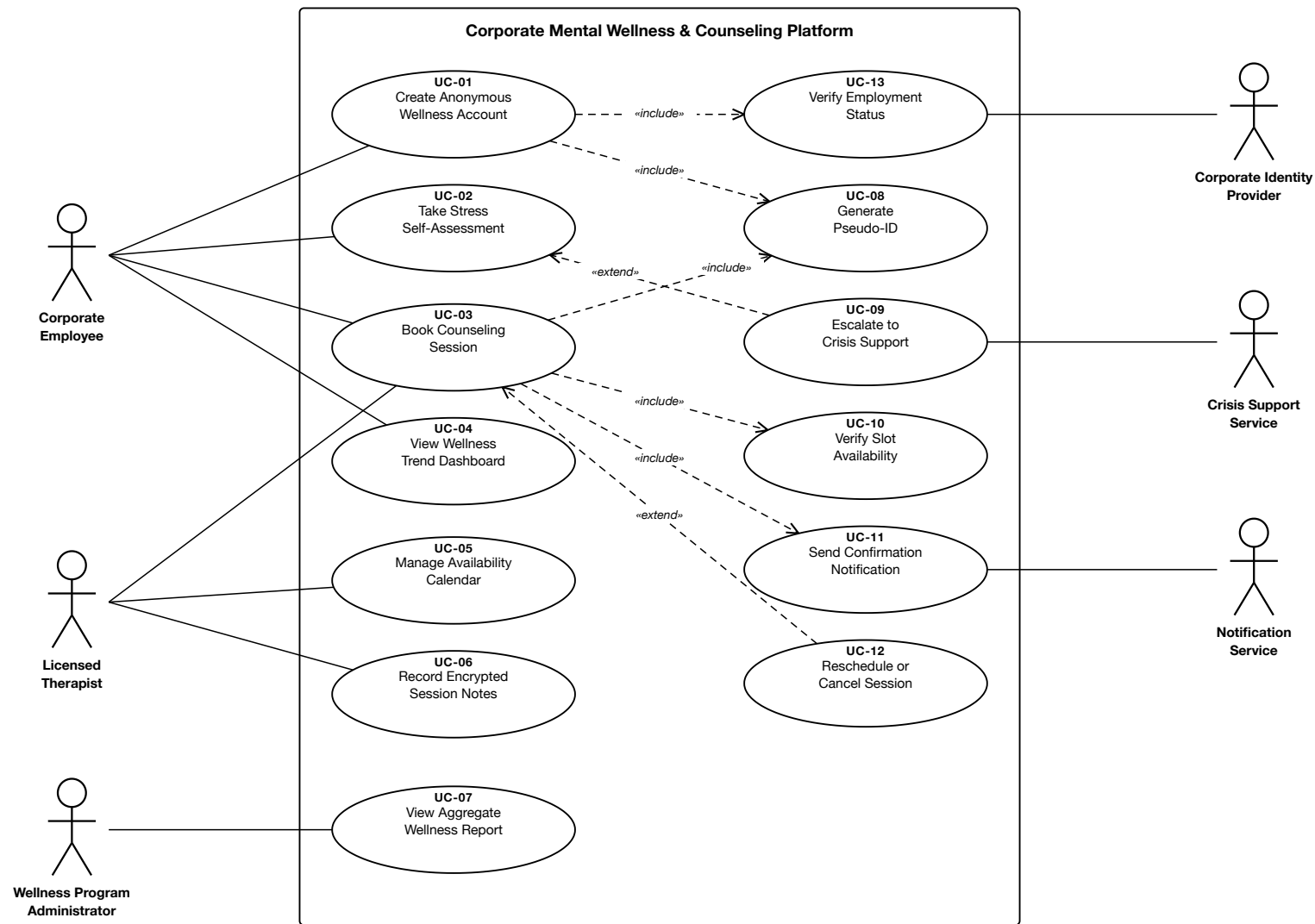
5. Use cases and traceability

Use case	Title	Actor / relationship	Traces to
UC-01	Create Anonymous Wellness Account	Corporate Employee	FR-001
UC-02	Take Stress Self-Assessment	Corporate Employee	FR-002
UC-03	Book Counseling Session	Corporate Employee, Licensed Therapist	FR-003
UC-04	View Wellness Trend Dashboard	Corporate Employee	FR-004
UC-05	Manage Availability Calendar	Licensed Therapist	FR-003
UC-06	Record Encrypted Session Notes	Licensed Therapist	NFR-002
UC-07	View Aggregate Wellness Report	Wellness Program Administrator	NFR-002
UC-08	Generate Pseudo-ID	included by UC-01 and UC-03	FR-001
UC-09	Escalate to Crisis Support	extends UC-02	FR-005
UC-10	Verify Slot Availability	included by UC-03	FR-003
UC-11	Send Confirmation Notification	included by UC-03	FR-003, NFR-001
UC-12	Reschedule or Cancel Session	extends UC-03	FR-003
UC-13	Verify Employment Status	included by UC-01	FR-001

6. Include and extend relationships

Relationship	Why it is modelled this way
UC-01 «include» UC-13	Registration always checks the person against the corporate directory first. An account that skipped the check would make the pseudo-ID meaningless.
UC-01 «include» UC-08	Account creation always generates a pseudo-ID; there is no account without one.
UC-03 «include» UC-08	Every booking is written against a pseudo-ID, so the identifier is resolved on each booking as well.
UC-03 «include» UC-10	A booking always checks the slot is still free before it is committed.
UC-03 «include» UC-11	A confirmed booking always sends a confirmation to both parties.
UC-09 «extend» UC-02	Crisis escalation runs only when the assessment scores in the severe band.
UC-12 «extend» UC-03	Rescheduling or cancelling happens only if the employee or therapist later changes the booking.

7. Use-case diagram



Lab 1 - Use-Case Diagram - Problem Statement #14 - Corporate Mental Wellness & Counseling Platform

8. Use-case flows

8.1 UC-01 Create Anonymous Wellness Account

Primary actor: Corporate Employee

Secondary actors: Corporate Identity Provider

Goal: Get the employee an account that proves they are entitled to the benefit, while leaving the counselling side of the system with nothing but a pseudo-ID.

Trigger: The employee opens the portal for the first time and chooses 'Create account'.

Preconditions

1. The employer has enrolled in the programme and its directory feed is reachable.
2. The employee holds valid corporate credentials.
3. No wellness account exists yet for this employee.

Postconditions (success)

1. A pseudo-ID in the format WEL-XXXXXXX has been generated at random and issued.
2. The mapping between the employee and the pseudo-ID is stored encrypted, in the identity store rather than with counselling data (NFR-002).
3. The employee can take an assessment and book sessions.
4. No counselling-side record created by this use case carries a corporate identifier (FR-001).

Postcondition (failure): No account, no pseudo-ID and no mapping row exist, and the employee is told the verification did not succeed.

Main success scenario

1. The employee chooses 'Create account'.
2. The system hands the employee to the Corporate Identity Provider to sign in.
3. The employee authenticates with their corporate credentials.
4. The identity provider returns a signed assertion confirming current employment («include» UC-13).
5. The system checks that no wellness account already exists for that assertion.
6. The system generates a random pseudo-ID («include» UC-08) and confirms it is not already in use.
7. The system writes the mapping to the identity store, encrypted.
8. The system drops the corporate identifiers from the session and keeps only the pseudo-ID.
9. The system shows the employee their pseudo-ID and explains that this is the only thing a therapist will ever see.
10. The use case ends successfully.

Alternate flows

4a. The person is not a current employee

- 4a1. The identity provider returns no assertion, or one marking the account as inactive.
- 4a2. The system refuses registration and shows the employer's benefits contact.
- 4a3. Nothing is written to either store and the use case ends.

5a. An account already exists

- 5a1. The system finds a mapping for that assertion.
- 5a2. The system signs the employee into the existing account instead of issuing a second pseudo-ID.
- 5a3. The use case ends successfully, with the pseudo-ID unchanged.

7a. The mapping cannot be written

- 7a1. The identity store rejects the write or times out.
- 7a2. The system discards the generated pseudo-ID so no orphan identifier is left behind, and asks the employee to try again.

Notes

- Steps 3 and 4 are the only point in the whole system where a corporate identity is handled. Everything after step 8 works on the pseudo-ID.
- The pseudo-ID is random rather than a hash of the employee ID, so nobody can recompute it from a directory export.

8.2 UC-02 Take Stress Self-Assessment

Primary actor: Corporate Employee

Secondary actors: Crisis Support Service (severe results only)

Goal: Give the employee a current reading of their stress level and point them towards help when the score is high.

Trigger: The employee chooses 'Check my stress level'.

Preconditions

1. The employee is signed in and the session carries only their pseudo-ID.
2. A published version of the ten-item questionnaire is active.

Postconditions (success)

1. The answers and the resulting score are stored encrypted against the pseudo-ID.
2. The employee has seen their score out of 40 and its band.
3. The assessment is available to the trend in UC-04.
4. Where the band is severe, helpline details and a priority slot have been offered (FR-005).

Postcondition (failure): Nothing is stored, the employee's previous assessments are untouched, and no band is reported.

Main success scenario

1. The employee chooses 'Check my stress level'.
2. The system presents the ten items, each with the five-point frequency scale from Never (0) to Very Often (4).
3. The employee answers every item.
4. The employee submits the questionnaire.
5. The system rejects the submission if any item is unanswered, naming the items that are missing.
6. The system totals the answers to a score out of 40 and maps it to its band.
7. The system stores the answers and the score against the pseudo-ID, encrypted (NFR-002).
8. The system shows the score and band, stating plainly that this is a wellness check and not a medical diagnosis.
9. The system offers a link to book a counselling session (UC-03).
10. The use case ends successfully.

Alternate flows

6a. The score falls in the severe band («extend» UC-09)

- 6a1. The system shows regional crisis helpline details above the score.
- 6a2. The system offers the earliest priority appointment slot starting within 24 hours.
- 6a3. The employee either takes the priority slot, which continues into UC-03, or dismisses the panel; either way the helpline stays visible on the result.

3a. The employee leaves partway through

- 3a1. The employee closes the questionnaire before answering every item.
- 3a2. The system discards the partial answers rather than saving a draft, so an incomplete set is never scored or plotted.

Notes

- The banding is fixed and published, so the same score always produces the same band and the result is reproducible.
- Step 5 is what keeps FR-002's fail condition, scoring a partly answered form, from happening.

8.3 UC-03 Book Counseling Session

Primary actor: Corporate Employee

Secondary actors: Licensed Therapist (indirect), Notification Service

Goal: Reserve a confidential counselling slot with a licensed therapist without revealing the employee's corporate identity.

Trigger: The employee chooses 'Book a session' from the portal.

Preconditions

1. The employee has an active account and a pseudo-ID has been issued (UC-01).
2. The employee is signed in, and the session carries only the pseudo-ID.
3. At least one licensed therapist has published availability (UC-05).

Postconditions (success)

1. A booking record links the pseudo-ID, the therapist and the chosen slot.
2. The slot is marked unavailable and cannot be booked by anyone else.
3. Both parties hold a confirmation showing the time in their own timezone (NFR-001).
4. The therapist's view of the booking shows the pseudo-ID only (FR-001).

Postcondition (failure): No slot is reserved, the therapist's availability is unchanged, and the employee is told why the booking did not complete.

Main success scenario

1. The employee selects 'Book a session'.
2. The system resolves the signed-in account to its pseudo-ID («include» UC-08).
3. The employee chooses a counselling area and a preferred date range.
4. The system retrieves matching published slots and displays them converted to the employee's local timezone.
5. The employee selects a slot and a session mode (video or chat).
6. The system re-checks that the slot is still free («include» UC-10).
7. The system reserves the slot and writes the booking against the pseudo-ID inside a single transaction.
8. The system generates the session join link and stores it against the booking.
9. The system sends confirmations to the employee and the therapist («include» UC-11), each in the recipient's own timezone.
10. The system shows the employee a confirmation screen with the pseudo-ID, the slot and the join link.
11. The use case ends successfully.

Alternate flows

4a. No therapist has availability in the requested range

- 4a1. The system reports that no slots match and shows the earliest date that does have availability.
- 4a2. The employee either widens the date range and returns to step 3, or leaves the booking; the use case ends without a booking.

6a. The slot was taken while the employee was choosing

- 6a1. The availability check finds the slot already reserved.
- 6a2. The system tells the employee the slot is no longer available and does not reserve anything.
- 6a3. The system refreshes the slot list and returns the employee to step 5.
- 6a4. If the employee's next two attempts also clash, the system offers to place them on the therapist's waitlist and the use case ends without a booking.

9a. The notification service is unreachable

- 9a1. The booking is already committed, so the system keeps it and queues the confirmation for retry.
- 9a2. The confirmation screen at step 10 warns that the email is delayed and shows the join link on screen.
- 9a3. The system retries delivery for up to one hour, then raises an alert for support.

Notes

- Nothing written by this use case carries a corporate identifier. Every record is keyed on the pseudo-ID (FR-001).
- Steps 4 and 9 are where timezone conversion matters, so they are the steps NFR-001 is measured on.
- Once a booking exists, UC-12 extends this use case to reschedule or cancel it.

8.4 UC-04 View Wellness Trend Dashboard

Primary actor: Corporate Employee

Secondary actors: None

Goal: Let the employee see how their own indicators have moved across recent assessments.

Trigger: The employee opens 'My wellness' from the portal.

Preconditions

1. The employee is signed in and the session carries only their pseudo-ID.
2. At least one assessment has been completed (UC-02).

Postconditions (success)

1. The employee has seen their stress score, sleep hours and sessions attended for up to their six most recent assessments, in date order.
2. Nothing has been written. The use case only reads.
3. No other employee's data was retrieved at any point (FR-004).

Postcondition (failure): No chart is shown and the employee is told the dashboard could not be loaded. No stored assessment is altered.

Main success scenario

1. The employee opens 'My wellness'.
2. The system fetches the assessments held against that pseudo-ID and no other, taking the six most recent.
3. The system decrypts the stored values for display (NFR-002).
4. The system plots stress score, sleep hours and sessions attended against the assessment dates, oldest first.
5. The system leaves dates with no assessment blank instead of estimating a value.
6. The employee selects a point to see that assessment's score and band.
7. The use case ends successfully.

Alternate flows

2a. The employee has no assessments yet

- 2a1. The system shows an empty dashboard explaining that a trend appears after the first assessment.
- 2a2. The system offers a link to UC-02 and the use case ends.

2b. Fewer than six assessments exist

- 2b1. The system plots only the assessments that exist.
- 2b2. The axis stops at the earliest real assessment rather than padding the chart out to six points.

Notes

- The scope of the query in step 2 is what FR-004's fail condition tests. A dashboard that pulled another pseudo-ID's rows would fail.
- Refusing to interpolate in step 5 matters because an invented data point would misrepresent somebody's mental health history.